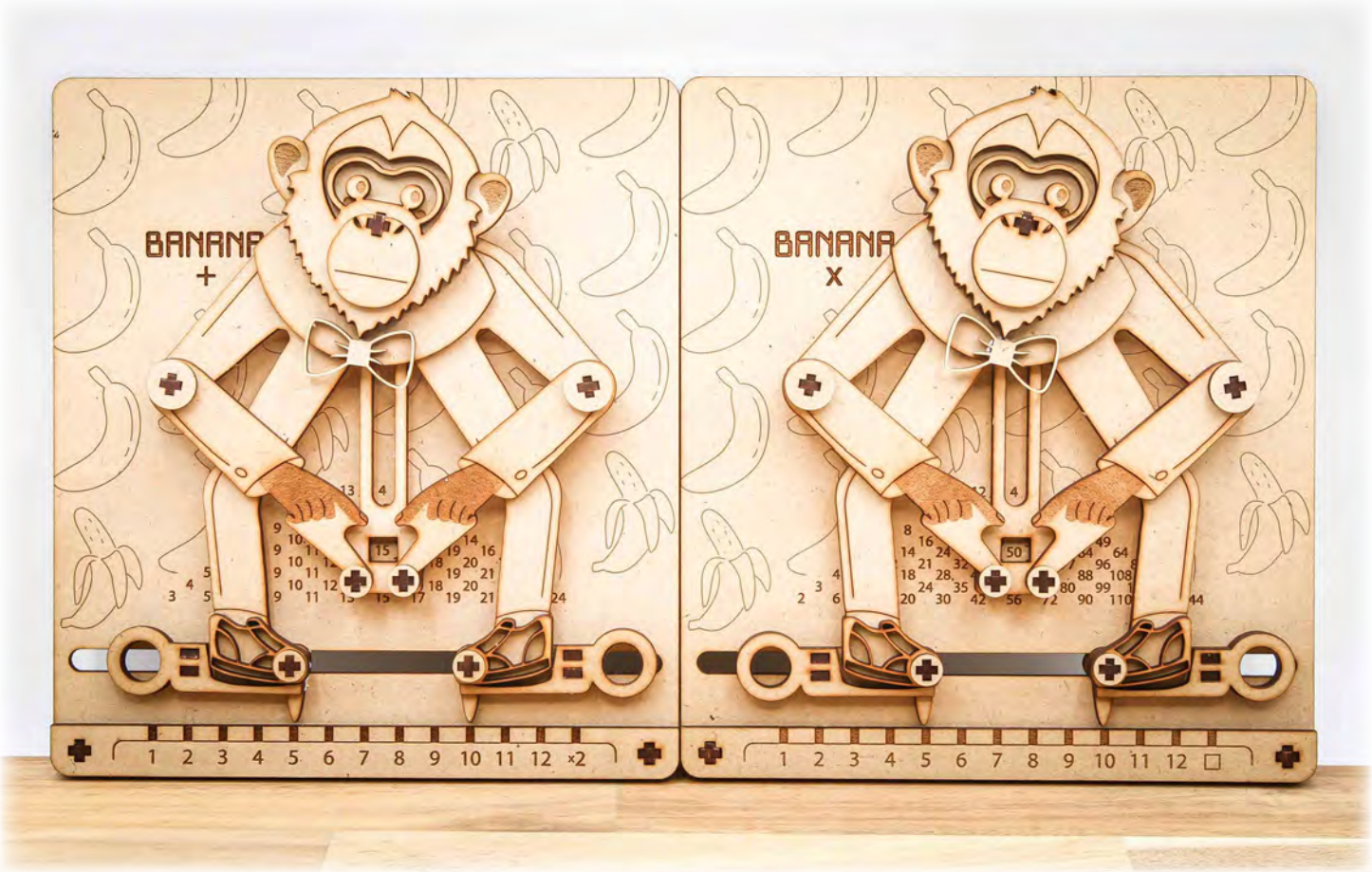
Mechanical Monkey Calculator

Multiplication and Addition

based on 'Consul The Educated Monkey' from 1916

by Ross McSweeney



Mechanical Monkey Calculator Copyright © 2024 by Ross McSweeney

made by Ross

This model is based on 'Consul the Educated Monkey' which was invented by the National Cash Register draftsman William Henry Robertson in 1916. When the arrows under the monkey's feet are pointing at two numbers, its fingers locate their product. I have also included an addition version! The original model was made from thin sheets of tin. My model is made from wood so required a bit of reworking to make it all work with thicker material. I have also updated the look to be a bit more friendly and welcoming than the original model. The monkey still retains his suit and the mechanism is as close to the original as I could possibly get it in wood format.

Important information

Thank you very much for purchasing these plans. You have my absolute thanks. Every time someone purchases a file from me it gives me more motivation to work on the next model. So thank you again, it honestly means a lot to me! Please feel free to contact me anytime if you have questions, need different file formats, dimensions, requests or need help with something. Please give me at least 12 hours to reply to make up for our various time differences. The first 4 pages of these instructions are full of important information so please read through it once if possible!

You can reach me at rossmcsweeney@gmail.com

My collection of automata and laser cut projects continues to grow thanks to your support. To see a full list of my models you can see them all on my Etsy page, Instagram page or Youtube page.

https://www.etsy.com/shop/RossMcSweeney

https://www.instagram.com/ross.mcsweeney

https://www.youtube.com/rossmcsweeney

made by Ross

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DO NOT DISTRIBUTE OR SELL THESE DIGITAL FILES

If these files end up getting shared around online I will stop making models this way and begin making physical kits only which wouldn't be as much fun for us all.

If you have purchased a random set online with this file included in the collection, then this file has been stolen. Please do not support piracy.

If you need any help with anything at all, size changes, design changes, extra file formats etc or you have some ideas or requests for some new models then feel free to email me at rossmcsweeney@gmail.com

Please give me at least 12 hours to reply to compensate for various time differences.

All the best with everything you are doing

Take care
Ross McSweeney

A note about Cutting order/Power settings etc

I have made different colours in the files to help your laser cutter cut things correctly. Vector engrave means that the laser draws the pictures, much like you would do with a pencil so the speeds and power settings are much lower. Raster engrave means that the laser moves left to right, line by line, much like how a printer works. You can go much faster with this type of engraving.

1 - **Purple** - Vector Engrave **Power 15% Speed 20 mm/s**
2 - **Green** - Raster Engrave **Power 30% Speed 150 mm/s**
3 - **Blue** - Raster Engrave **Power 30% Speed 100 mm/s**
4 - **Yellow** - Vector Engrave **Power 20% Speed 20 mm/s**
5 - **Pink** - Cut **Power 35% Speed 10 mm/s**
6 - **Black** - Cut **Power 35% Speed 10 mm/s**

I use a 40W laser cutter to cut all my pieces. I use very low power settings and slow speeds to save the life of my laser tube.
My settings will differ to yours but I will write what I use just in case it helps someone.
I strongly recommend that you engrave/cut in this order if possible. If you have the wood sitting directly on the laser bed then the cutting order doesn't matter at all. If you have the pieces raised off the laser bed this will stop them falling down at the wrong time. Raising the pieces off the laser bed also stops the back of the wood getting burn marks.

Purple is for the banana pattern. Just skip this colour if you don't want the bananas. I engrave this part very, very lightly as I don't want the bananas to be too obvious.
Green is for the part numbers, some details and also adds colour to the hands.
Blue is for the number markings. This could actually just be the same as green but I like to engrave these pieces a bit deeper. There is no reason to do this, I just thought it looked good.
Yellow is for adding some detail, like on the suit and around the hands. I like to do this after all the other engraving is complete to make the lines nice and crisp.
Pink gets cut first. Pink is usually found within a black piece. Most laser cutters should start from the inside of a piece but just in case yours doesn't, I made them 2 different colours.
Black gets cut last.

What you will need - Glue, toothpicks and candles

I make all my models using only pieces that you will cut yourself. Gluing some pieces saves so much hassle and you can safely let anyone play with the models. I will make note of places where I think you will need glue. Use a toothpick to help get glue in all the tiny places. I also recommend getting a white candle to wax the moving parts. I use a tea light from the dollar store.

Wax makes this model slide beautifully so I highly recommend using it!

Importing Issues

If you are using an older version of **CorelDraw** you may see some crazy warped lines. This only happens in some of the older versions of the software. If you see any crazy lines please send me a message with what version you are using and I can resave the files for that particular version of Corel Draw.
I also have Illustrator files if anyone needs them.

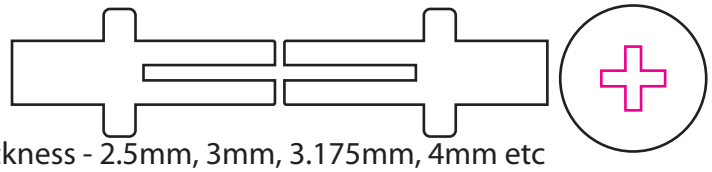
I 100% recommend cutting the test file first to make sure everything is good before you waste time and wood cutting everything out. I also recommend reading the instructions through once before starting.

Make sure your laser bed is perfectly level. Make sure your sheets of wood aren't warped.
This model relies on everything being cut precisely.

Good luck!

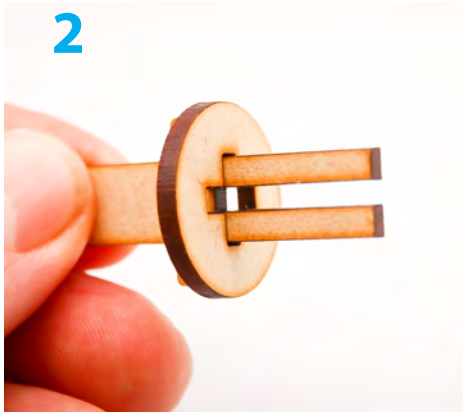
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The Test Cut



1 - Choose the test cut file that matches your wood thickness - 2.5mm, 3mm, 3.175mm, 4mm etc and cut it out. It is very small and quick to do. This will check your laser's kerf, wood thickness and tolerances etc.





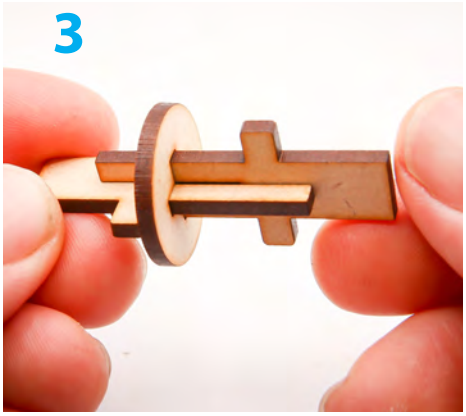
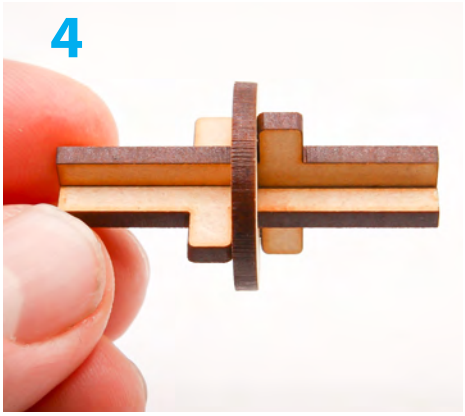
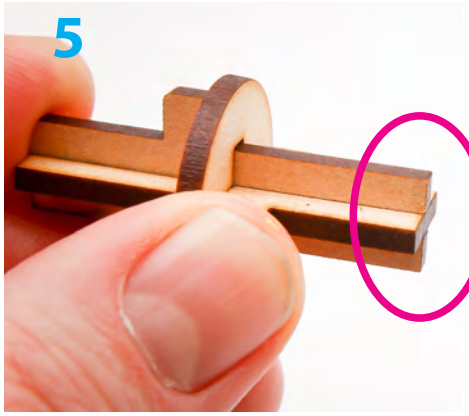
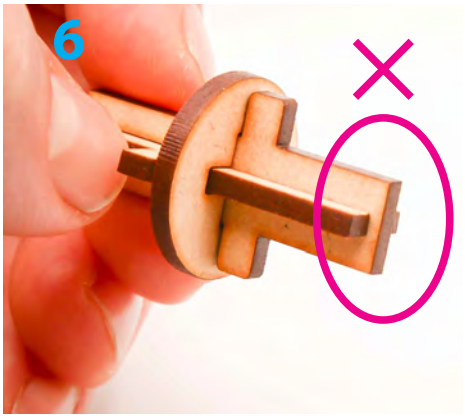


Photo 1: These are the test cut pieces. **Photo 2**: Thread the circle onto the legs of one shaft piece. It should slide on without being too tight or too loose. **Photo 3**: Now thread the other shaft piece in like I am doing in this photo. **Photo 4**: Slide it all the way together and that circle should be perfectly locked in the middle. **Photo 5**: The ends of the cross shaft should be perfectly flush. **Photo 6**: This is not good. You can see that the shaft is not sitting flush. It means that the wood thickness is not correct.







What happens if you can't assemble this test cut properly?

Check to make sure your laser bed is even and that the laser head is not too close or too far away from the sheet of wood otherwise the laser's beam will be uneven and your cuts wont be straight. You can probably check by looking at your cuts and seeing a slight angle on them. **If you can insert the shaft into the cross hole from one side but not from the other side then your cuts are not even.**

Check the bed, settings and laser height.

What about if they just absolutely do not go together?

Make sure that the wood you are using matches the file you are using. Measure your wood as sometimes there can be variations between what the manufacturer says and what the actual thickness is. I have had someone tell me there wood was 3mm but after measuring it was actually 3.2mm. It doesn't seem like much but it makes a huge difference. Please message me if you need help with anything!

What about if you think it is a bit too loose or a bit too tight?

You could try offsetting the lines slightly but that would take forever. The best thing to do would be to scale the entire model up or down by 0.1% or so to make the entire model slightly tighter or slightly looser. See the next page of these instructions about scaling.

If you need help with something please send me an email and I can probably sort it out for you. As usual please give me at least 12 hours to reply to make up for our time differences.

Important note about wood thickness and cutting size

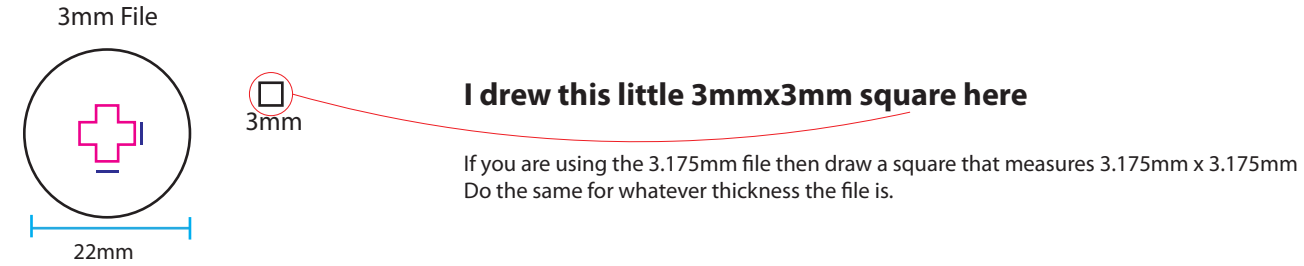
This model is designed with the wood I have locally available, which is 2.5mm MDF here in Japan. Everything in this model is designed to be made with this thickness of wood. However! You can change the size of the model to fit any thickness of wood. I have pre-made some different sizes in the download file. Read below for some explanations on how to make a new custom size.

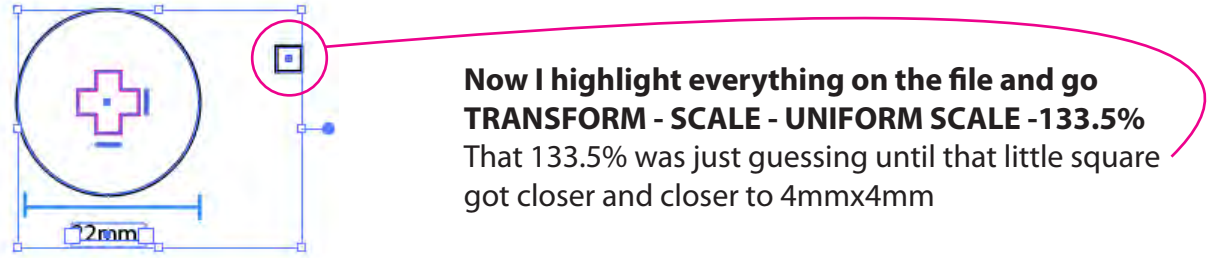
Redesigning the model for different thicknesses of wood is basically impossible. So instead, the simplest way would be to scale the whole model up to make these slots the same thickness as the wood you are using. This won't be a problem as long as the cutting dimensions of your laser cutter's bed are big enough to accommodate the resize.

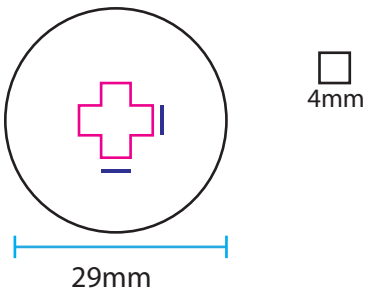
How to Scale - Open an SVG file in LightBurn, Inkscape, Illustrator or some other editing software.

For example - I want to make a file for wood that is 4mm thick. Open up any of the SVG files (I will use the 3mm file for this explanation)

Start by opening the 3mm SVG file, then draw a little square somewhere that measures 3mm by 3mm



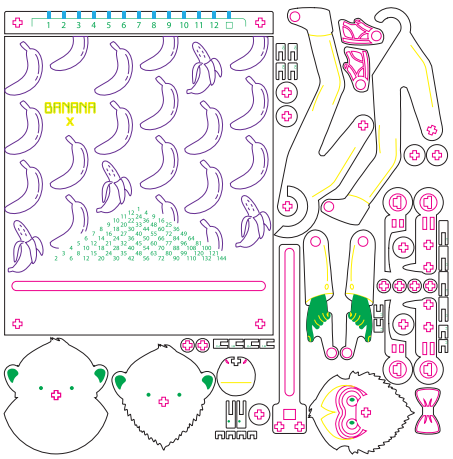


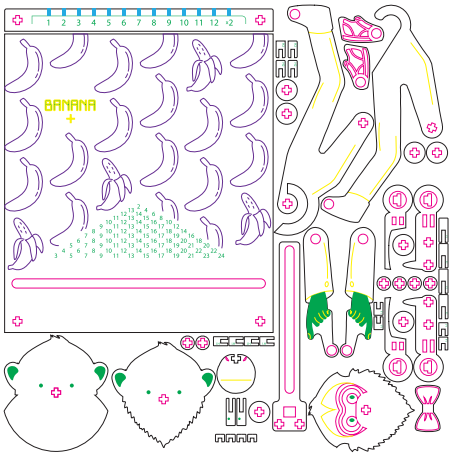


This is how I turned the original 2.5mm model into 3mm and 3.175mm versions. I can use this same method to make any size you like! I do not recommend going thinner than 2.5mm but it is possible.

And lastly

I recommend reading through the instructions once first so you kind of get an idea of what is going on. This is what the model looks like. On the left is all the parts for the multiplication version. On the right is the addition version. All of the parts are exactly the same except for the banana sheet and number strip. If you don't want to cut 2 monkeys then just make 1 monkey, then cut the board and number strips only and transfer your monkey between sheets.





Dimensions

The original cutting files are made to fit on these dimensions

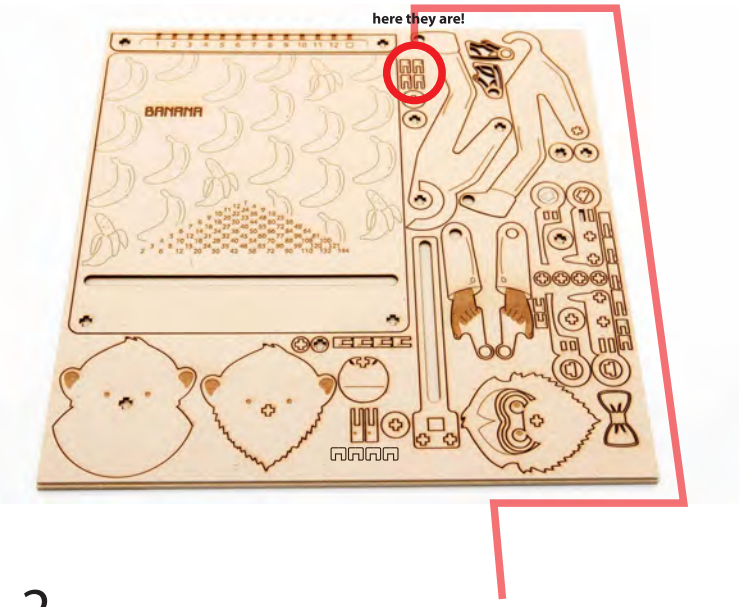
2.5mm - 30cm x 30cm	● (The banana board will be about 18cm x 20cm)
3mm - 36cm x 36cm	● (The banana board will be about 21cm x 24cm)
3.175mm - 38cm x 38cm	● (The banana board will be about 22cm x 25cm)
4mm - 48cm x 48cm	● (The banana board will be about 28cm x 31cm)

I will also try and make some different dimensions for those who can't cut the above dimensions.

At the time of writing these instructions I haven't made the extra files yet but they will probably be about 11x18" as that is the most requested dimensions I receive.

Now enough talking and onto making!

Mechanical Monkey Calculator - Part 1



1 - Engrave and cut out all the parts.
This photo on the left here is 2.5mm on 30cm x 30cm sheets of MDF.

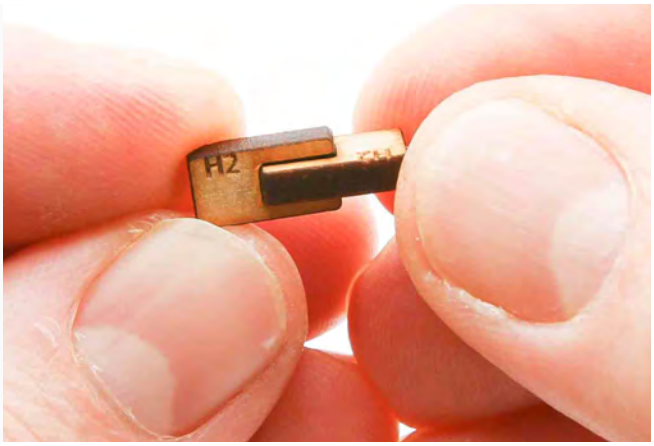
Note from future me: If you are going to paint your monkey it is easier to do it now rather than later.

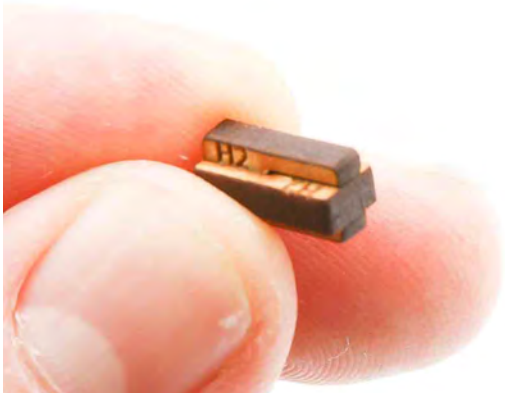
2 - Find the 4 little shaft pieces with H2 written on them.

Join 2 H1 pieces together as shown below to make a little cross shaft piece. Push them together and make sure that the ends are nice and flush. They should fit together tightly so you may need to press down hard on them to get them to slide together.

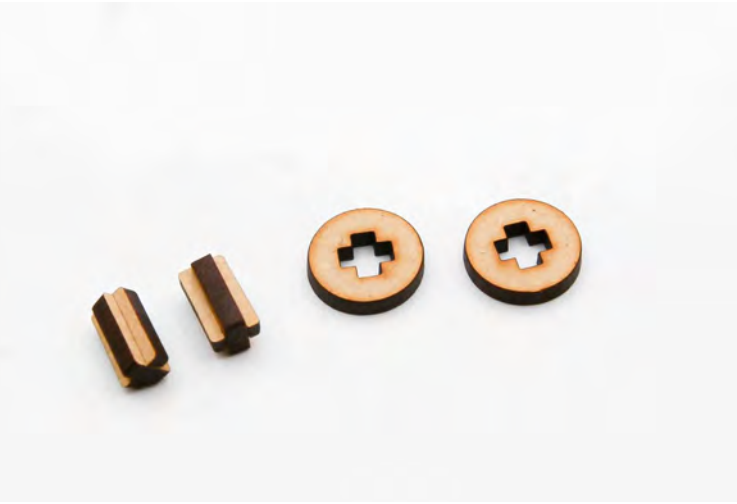
Make 2











3 - Find 2 of these large circle pieces with the cross holes in the middle. I think there are 7 scattered throughout the sheet but they are all the same size so don't worry about mixing them up.

4 - Add a tiny bit of glue to the inside of the cross hole. I use a tooth pick to add just the tiniest bit of glue in there. Push the shaft you just made into the cross hole and then push it down flat on a table or something so the shaft is sitting perfectly flush to the bottom and standing up straight. Or just make it look like the photo below if that is easier.





IMPORTANT **When using glue in this model, only use a tiny bit and make sure you don't get it where it shouldn't be as there are a lot of parts that need to move and slide perfectly.**

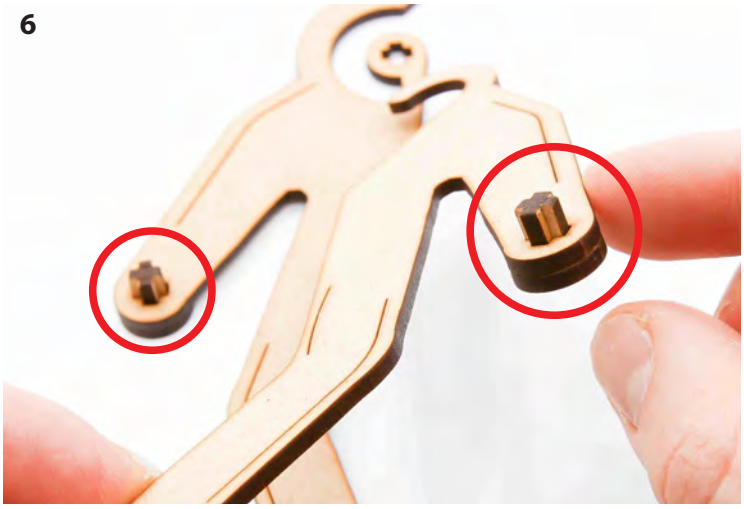


5 - Now get the 2 halves of the body.

Push the bits you just made into the arm slots from underneath. Or better still, push the arms down onto the shaft pieces you just made. If you find that they are a bit tight you can rub some wax on the shaft pieces to help them slide in super easily.

6 - Here they are both pushed in.

You don't need to glue them in. That circle stopper piece is just used to add some height to the body





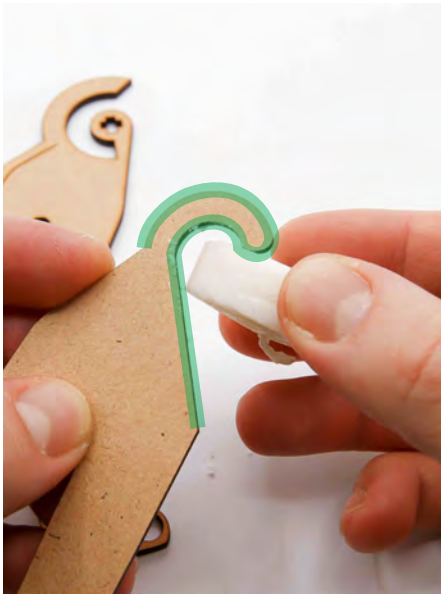
7 - Rub some wax around the shaft that is sticking out. Just a little bit is ok.

This will help turn the arms buttery smooth later on.

8 - This is probably the most important step of the whole model.

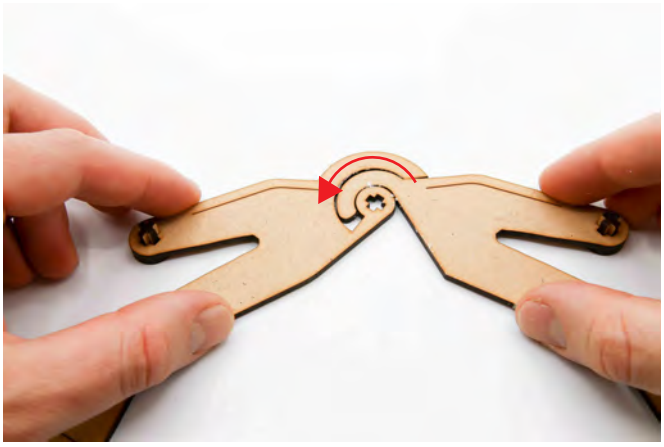
Rub some candle wax around these hook looking pieces on the body. On the next page you can see how they kind of work as a hinge with each other. We want the monkey to be able to scissor apart smoothly. So rub some wax against the edges as best as you can. If you can't get the candle in there just scrape the edges with the candle a little and rub it in with your finger a little bit. I highlighted the wax bits in green.





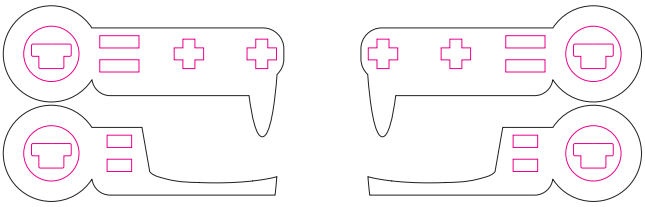






You need wax so the hinge will be able to slide like this. It should be very smooth. Do it a few times to work the wax in. I find that waxed wood always works much better a few hours later which is why we are doing it now. Put these pieces aside for now and we will come back to them later.

9 - Find these 4 pieces.



You can throw out all of the inner pieces except for this one.  Make sure you have 4 of them.

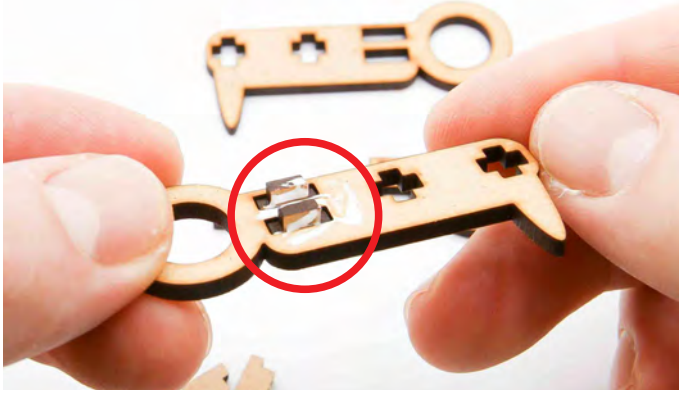
If you don't then they may have fallen through the bed of your laser cutter.

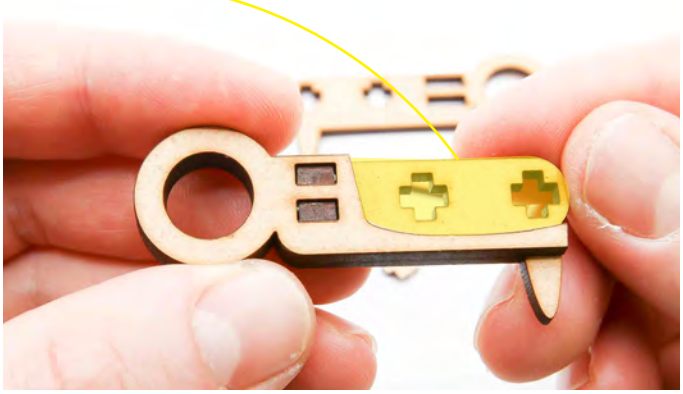
10 - Add a tiny drop of glue to those 2 long rectangular slots. (Pay attention to my orientation.) Now push those 2 little tab pieces down flush. The smaller end is sticking up.





11 - Add a bit more glue around those 2 tab pieces that are sticking out and then push on the top layer. Again pay attention to the orientation. Make sure yours looks exactly the same as my one on the bottom right here. Be careful that you didn't get any excess glue in this open space here.

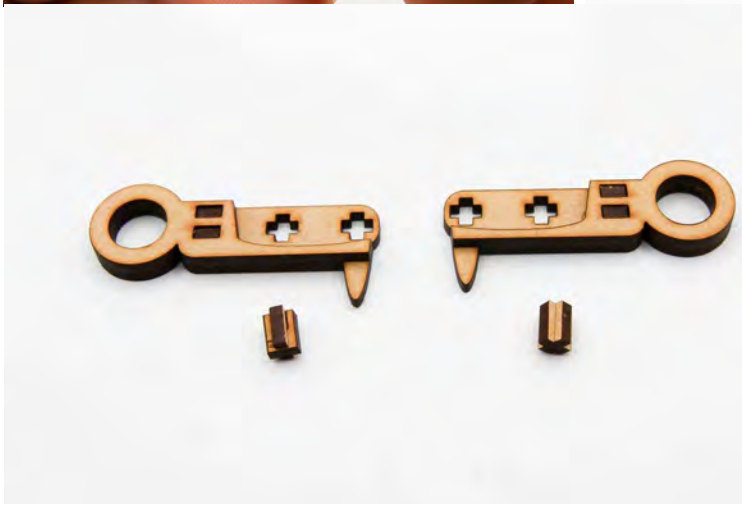


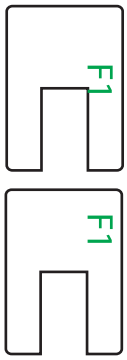


12 - Make sure that the back is nice and flush. Now you can make the other one and make sure that the orientation is exactly like this photo below on the right here.

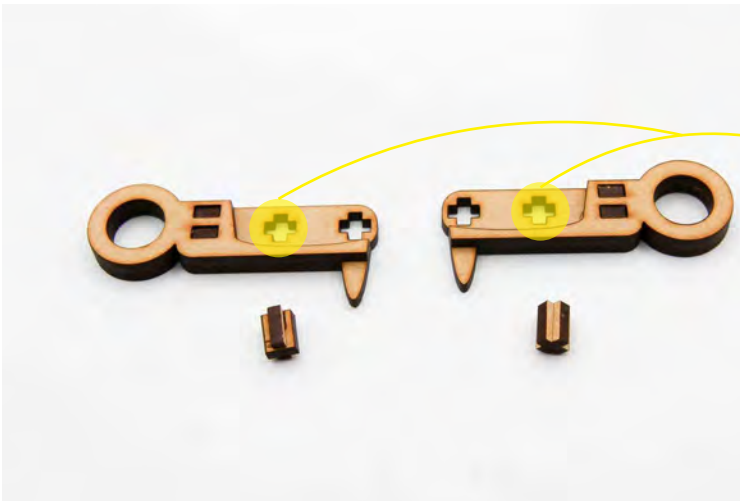








13 - Make 2 tiny F1 shafts. They go together like step 2 from before.



14 - Add a tiny drop of glue INSIDE the cross hole

here and here.

These little shafts will be glued in from the back. The shaft will be sticking out the back and the front will be flush.

Let me take some photos to make that easier to understand.



This cross hole gets the glue.



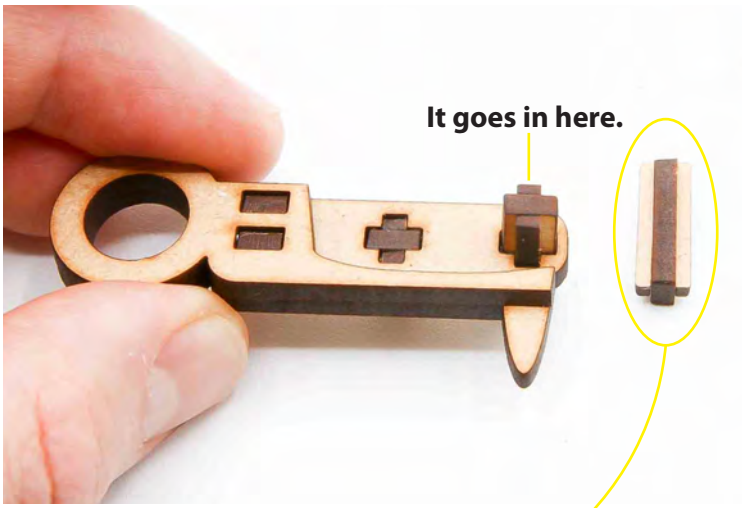


15 - Here you can see that the front is nice and flush, nothing is sticking up and there is no excess glue.

The other photo just here shoes the back. The shaft is sticking out nice and straight.



Here are both finished. Make sure you glued the shaft into the correct cross hole.





16 - Make the long F2 shaft the same way as before.

Rub some wax on it and then push it into the remaining hole.

Push it on roughly half way or until it is sticking out the back the same distance as the little shaft.

Don't worry too much about getting it perfect now as it will sort itself out later on.



17 - Make the other F2 shaft and push it into the other foot piece the same way.

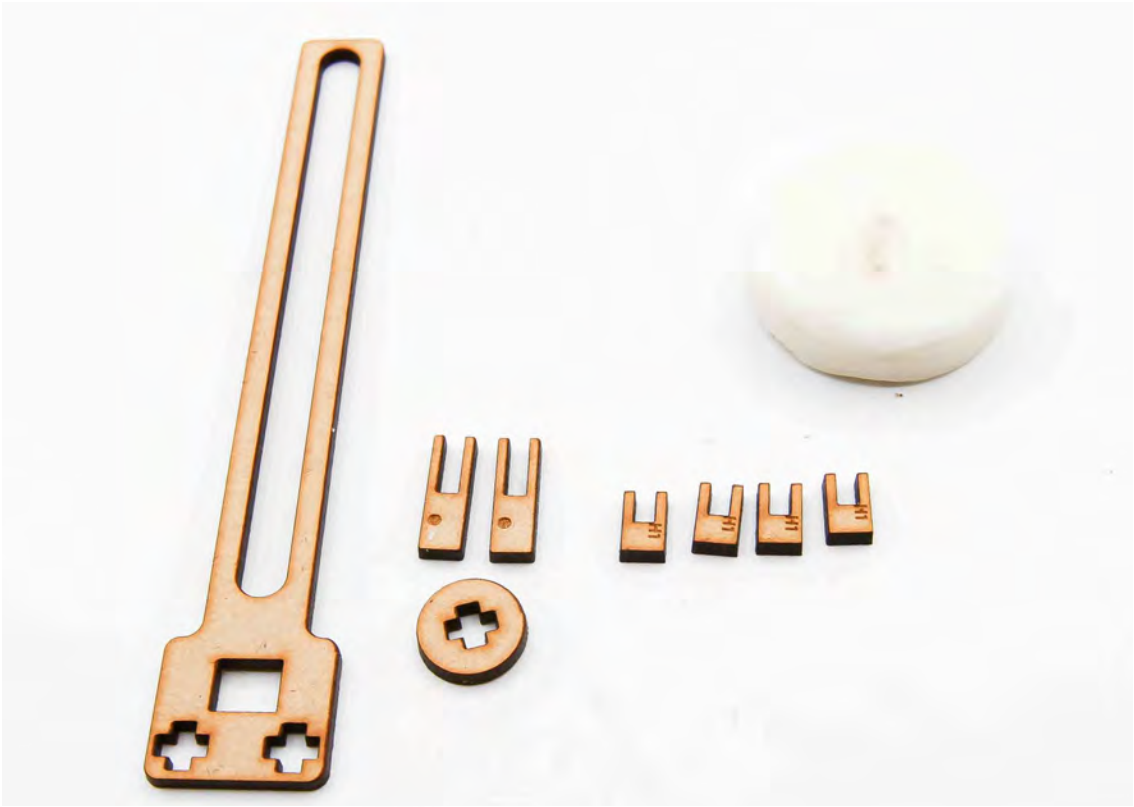
Rub some wax to make it slide easier, push it down about half way or preferably so it is the same height as the little shaft.

Now just put these pieces aside for later!

We are making a few little sub-assemblies before we put it all together.

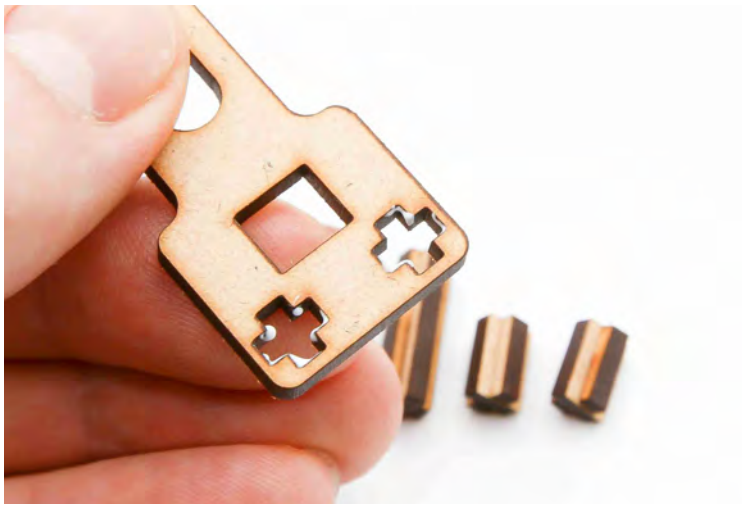
18 - Find all of these pieces shown below.

You need the long bar looking thing, the 2 long shaft pieces with the engraved circle on them, 1 of the large circle pieces, the 4 little shafts labeled H1 and some candle wax.





Start by assembling all the shaft pieces.

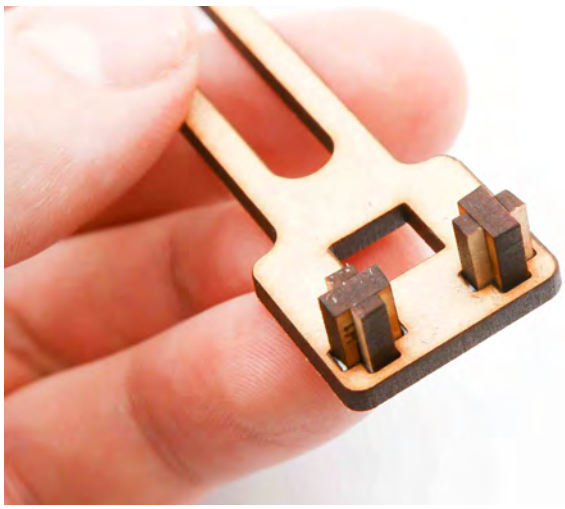


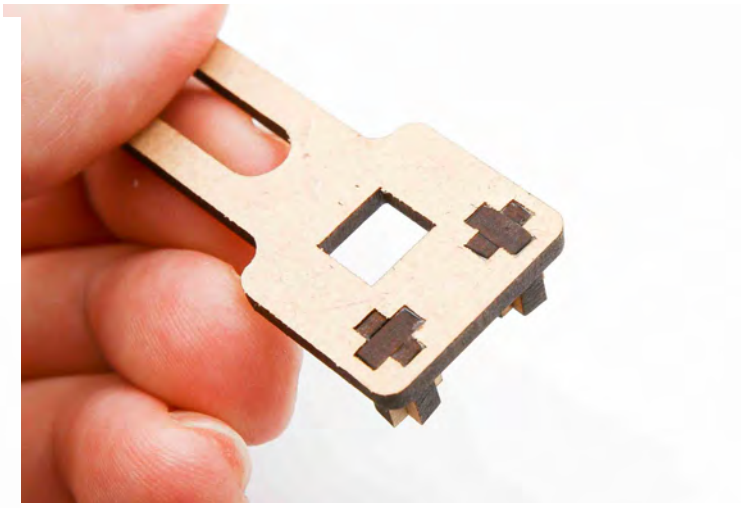
19 - Add a tiny drop inside those 2 cross holes.

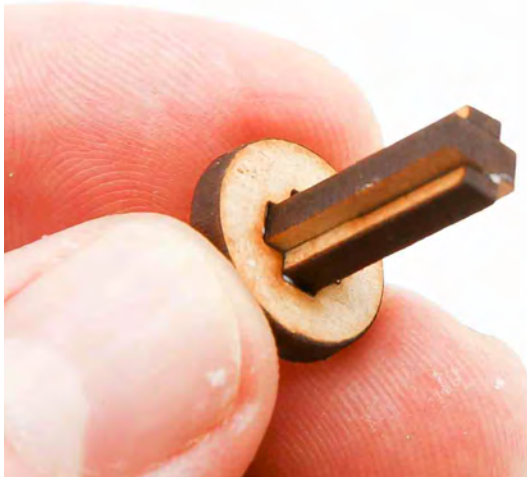
Then push the 2 tiny shafts into them.

Push them down flat on a table so that they are flush and sitting up perfectly straight.

If you flip the pieces over you can see that they are nice and flush.





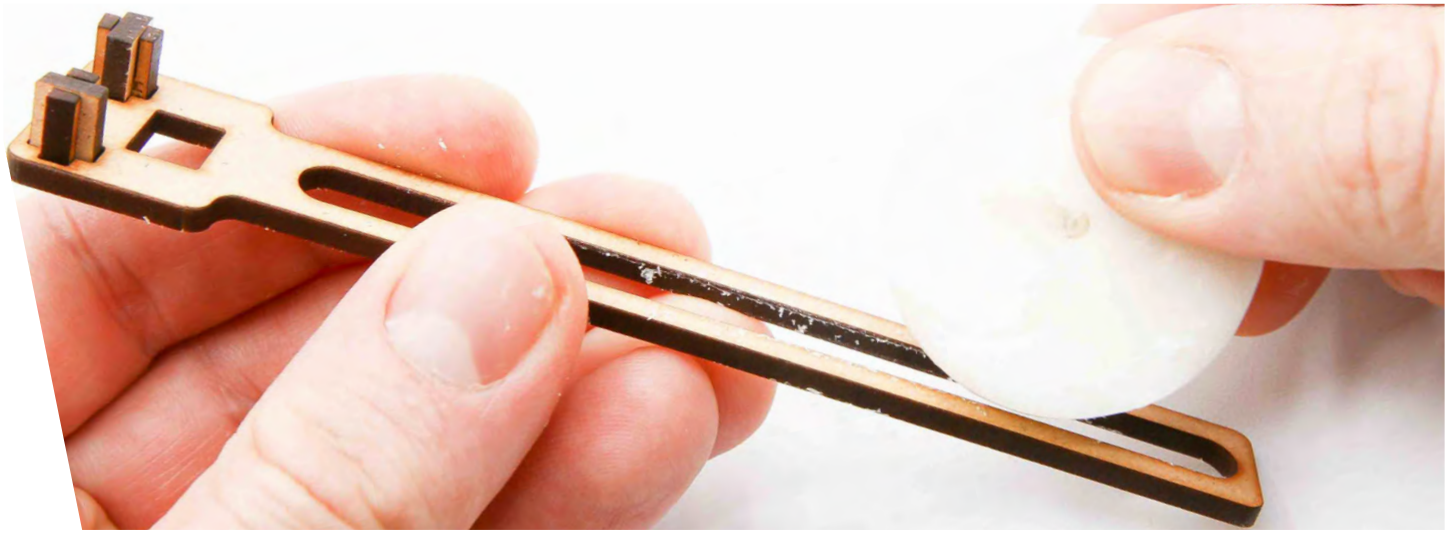


20 - You can now glue the long shaft with the engraved dots into the large circle piece.

Again, push it down on a table or something to make sure it is straight and flush.

Finally get some candle wax and rub it so the inside of this piece is nicely coated in wax.

This will help the monkey slide up and down.



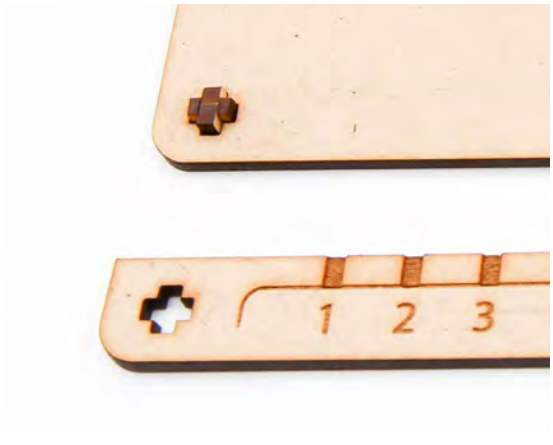


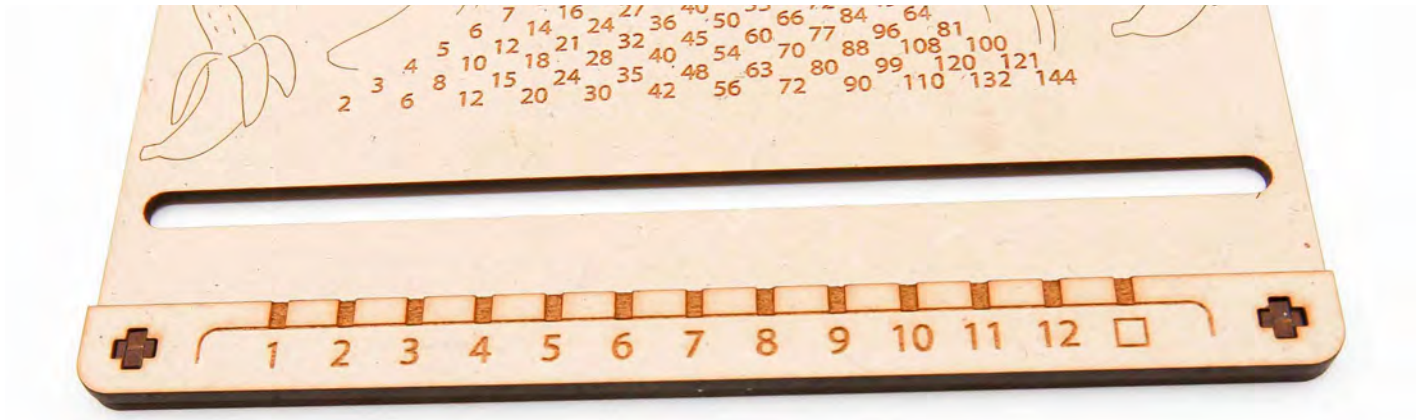
21 - Find these 4 incredibly tiny shaft pieces and assemble them to make 2 complete cross shafts.

Glue them into the banana base so that they are sticking up! Up is the side with all the engraving marks.

You can push the number strip onto the cross shafts now if you like but DON'T glue it on yet!

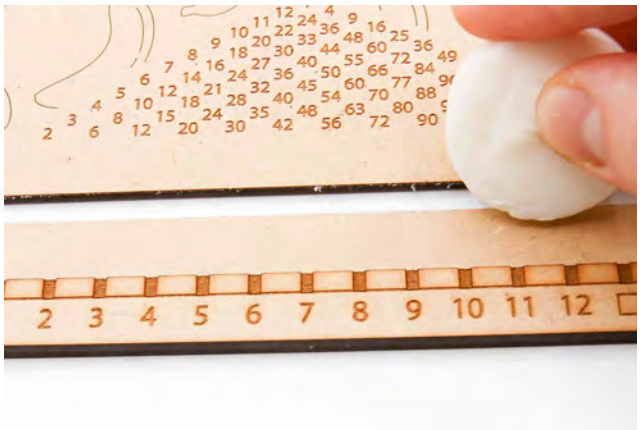






Rub some wax on both sides of this long hole. The monkey's feet are going to slide along here and it is good to make it slide as smooth as possible.



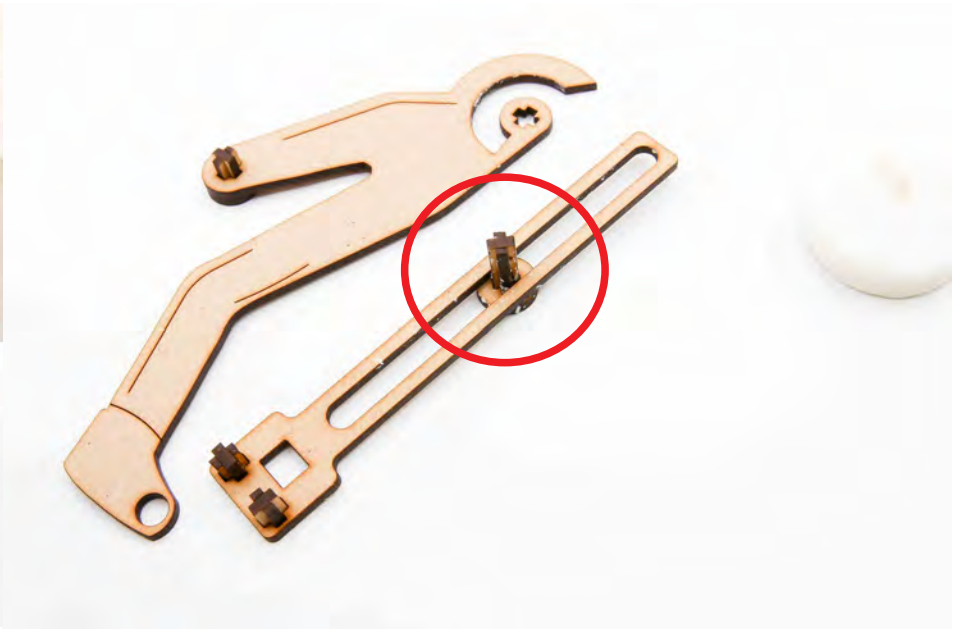


Part 2 - Assembly with bigger photos!

1 - Find these 3 parts that you have already finished.

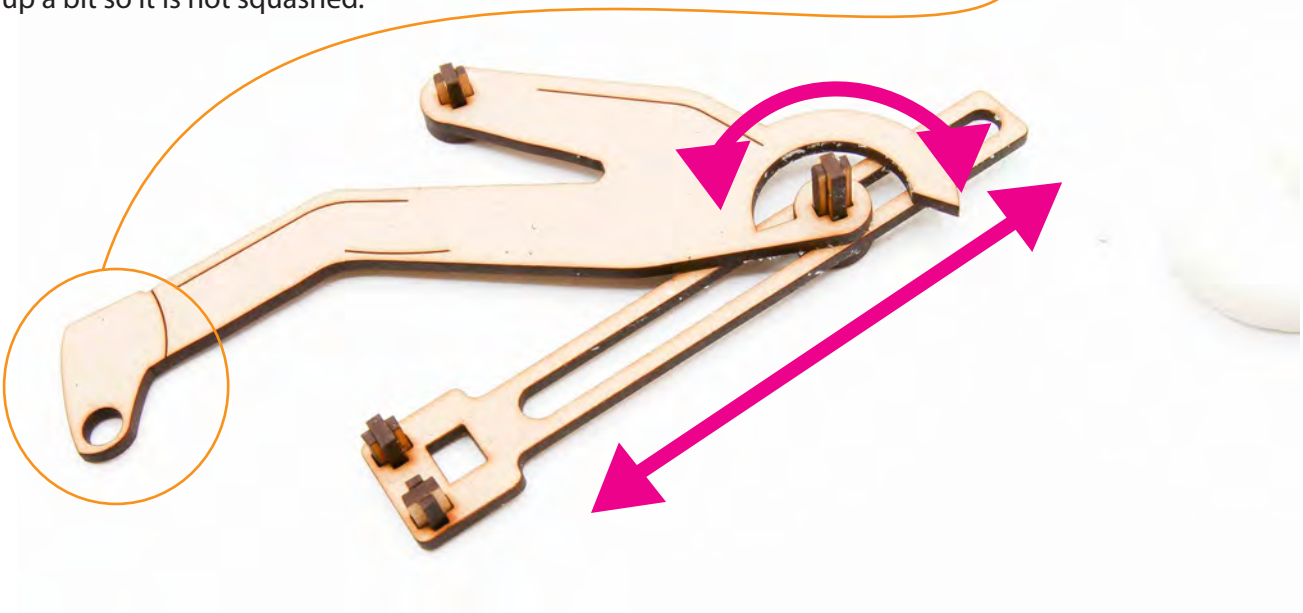






2 - Rub wax around that long shaft with the circle glued to the end. Now put the long arm with the slot cut out of it over the top of that shaft.... I'm sure there is an easier way to explain that...just make it look like the photo above.

3 - Now push this half of the monkey onto the cross hole. Be careful as that neck piece is a bit delicate. Push it all the way down **but not down tightly**. Make sure the foot is sticking out over this side of the middle piece and that the whole thing can slide and rotate freely. If it can't then try adding more wax or gently prizing the monkey back up a bit so it is not squashed.



It should be able to slide and rotate easily. Play with it for a little while until all the wax has worked its way into the wood.



4 - Gently rest the other half of the body like this. You can play with it a little more to make sure that it is still nice and loose.



5 - Let's make the head!

Push the body together like this.

Find that largest head piece and push it down onto that cross shaft piece. You can kind of see how that head piece will fit properly so I don't think you will get the orientation wrong.



6 - There it is on the body and here is the next piece waiting to go on! Exciting ~ but it looks nothing like a monkey yet.



7 - And here comes the next layer.

I should have mentioned this way earlier but if you are thinking about painting anything I recommend doing it before assembling it... I may go back and write a note about that now!

Done! I left a note at the beginning so now you already know to do this before I realised that I should have told you.



8- And here is the last layer.

Add a little bit of glue somewhere in the middle there and stick on his mouth/nose bit.



9- Look at that handsome monkey!

Make sure that the monkey can still slide up and down and rotate!

If it is too tight just pull the head off a little bit.



Here is something interesting!

Look at my nose shaft here... it is off by a few millimeters...
There are 7 layers of wood and I am using the 2.5mm file.
7 layers times 2.5mm is 17.5mm. The shaft I made is 18mm which is a bit longer to make sure that the monkey can still slide up and down freely.

So why is mine incredibly short?

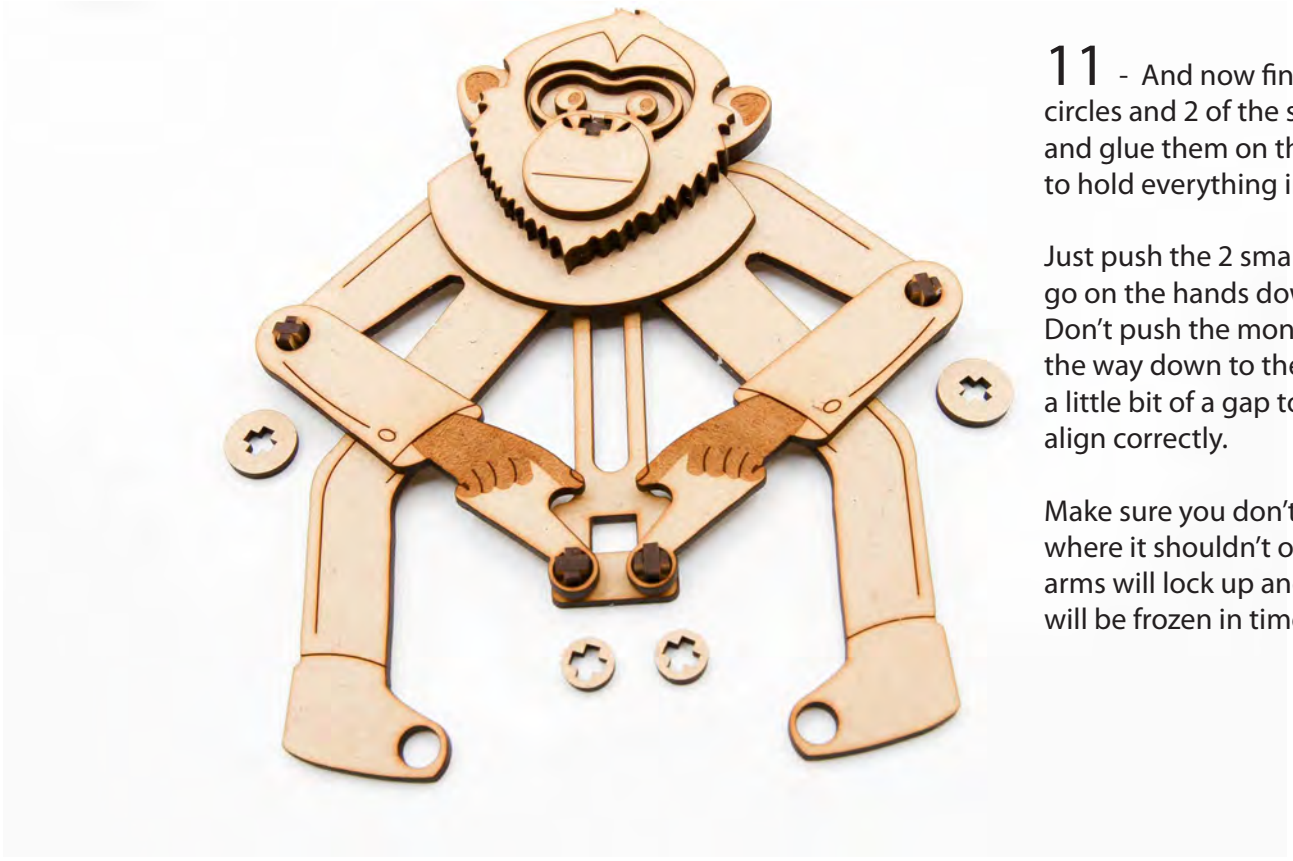
Recently I changed supplier as the new one was much cheaper and I have a sneaking suspicion that the new wood isn't 2.5mm. It would have helped for me to do my own test cut file before starting. This also explains why some of my pieces were a bit tighter... I thought it was the rain.

So what should I do? I could go back and remake the whole thing after resizing it, I could resize and recut just this shaft or I could leave it as is...

I feel like the model will still work fine, so this is a good lesson in making sure to always try the test cut file first! Onwards!



10- Get the 2 arms and line them up like this. The 2 pointing fingers are at the top.



11 - And now find 2 large circles and 2 of the small circles and glue them on the cross shafts to hold everything in place.

Just push the 2 small circles that go on the hands down gently. Don't push the monkeys hands all the way down to the bottom. I left a little bit of a gap to help things align correctly.

Make sure you don't get glue where it shouldn't otherwise the arms will lock up and your monkey will be frozen in time.



11.5 - Beauty shot!



12 - Put the monkey onto its 2 leg sliders as shown in the photo here.

You can see how the cut-outs match the foot contour.





13 - Add tiny dabs of glue and glue the shoes on making sure to keep to the outline of the foot.



13 - Add some of the the small circles to cap off the shoes.

Make sure that you don't push them down too tightly as the feet still need to be able to pivot.
Make sure that you don't get excess glue where it shouldn't be.



14 - This is over exaggerating but you can easily see that the feet are not glued to the feet sliders.

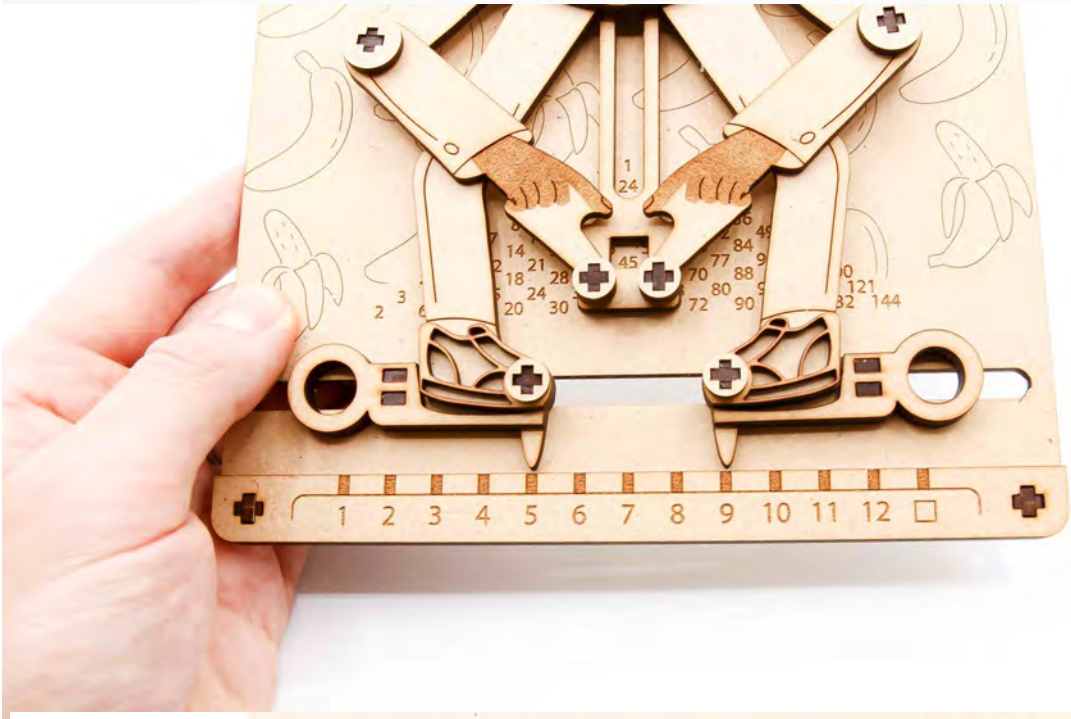
As we all found out before my wood is not the right thickness so I am just resting these stoppers half on. Normally you would push them down flush. But this is a good thing to learn! Just push them on enough so the feet don't fall off and the legs can move freely. I imagine you would normally push them on flush as I designed it but I have lost confidence in myself now...



Here is what the monkey looks like so far with all of it's stoppers on. I can't even notice that the stoppers aren't pushed on flush anyway.

I made these leg stoppers like this to stop the monkey tilting slightly from side to side and messing up the number position. Wood is much heavier than the original tin version!



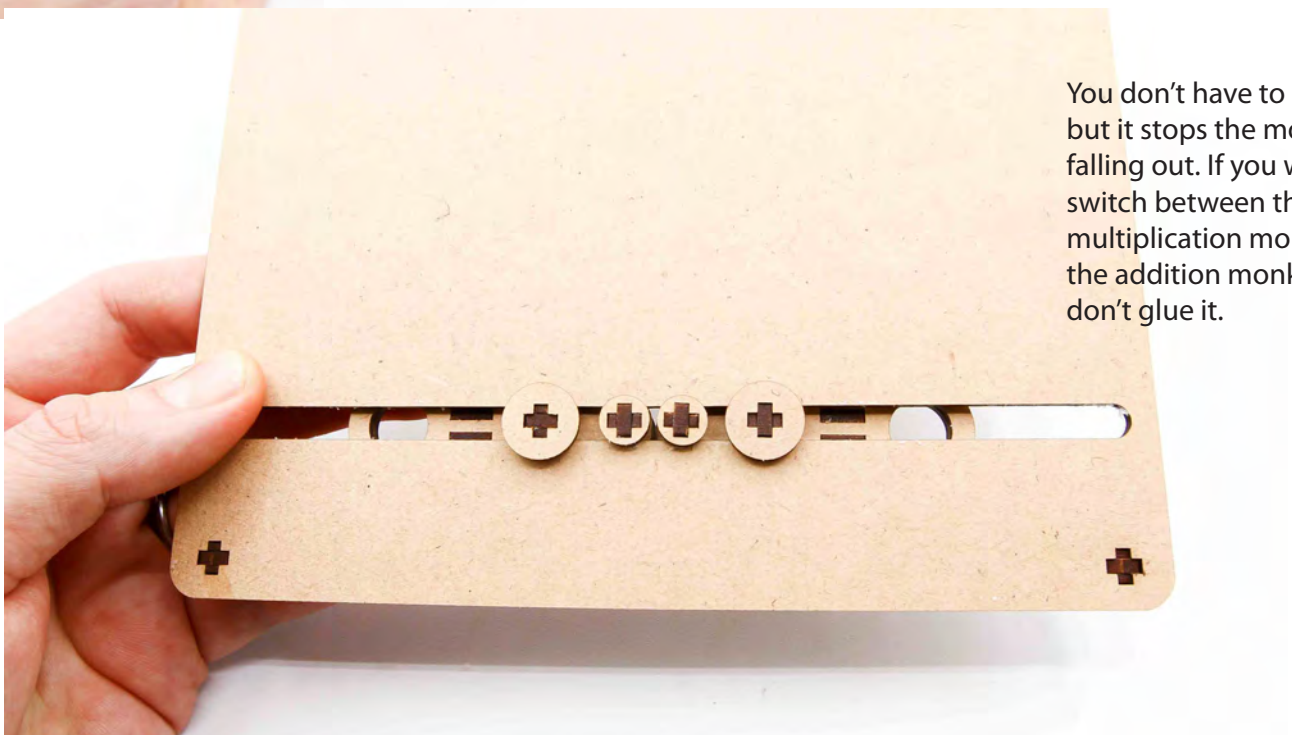


15 - And here is the last part!

Simply pop your monkey into that slot so both shaft pieces go through.

Then flip it over and you will see the 4 shaft pieces sticking out the other side.

Glue 2 large circles on the outside shafts and glue 2 small shafts on the inside shafts as shown in the photo below. Don't mess that up or the feet won't be able to get close to each other.



You don't have to glue them in but it stops the monkey from falling out. If you want to switch between the multiplication monkey and the addition monkey then don't glue it.

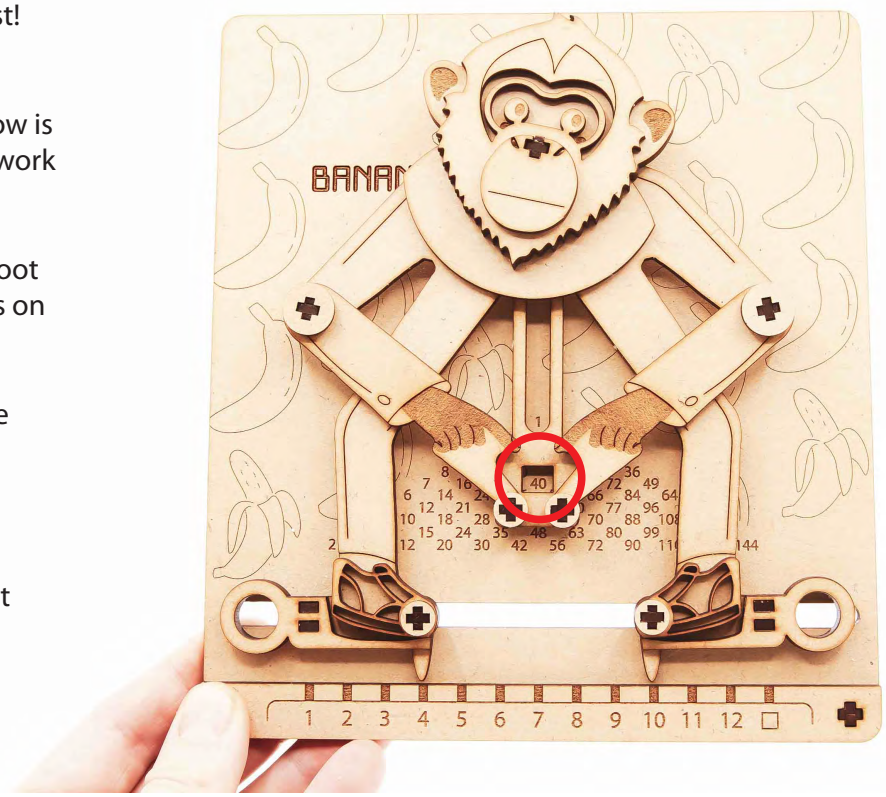
16 - Let's give him a test!

Slide the feet until the arrow is pointing to 2 numbers to work out the answer.

Here you can see the left foot is on 4 and the right foot is on 10.

If you look up to where the monkey is pointing, he is pointing straight to 40!

Success!!! Even with my incorrect wood thickness it still works beautifully.





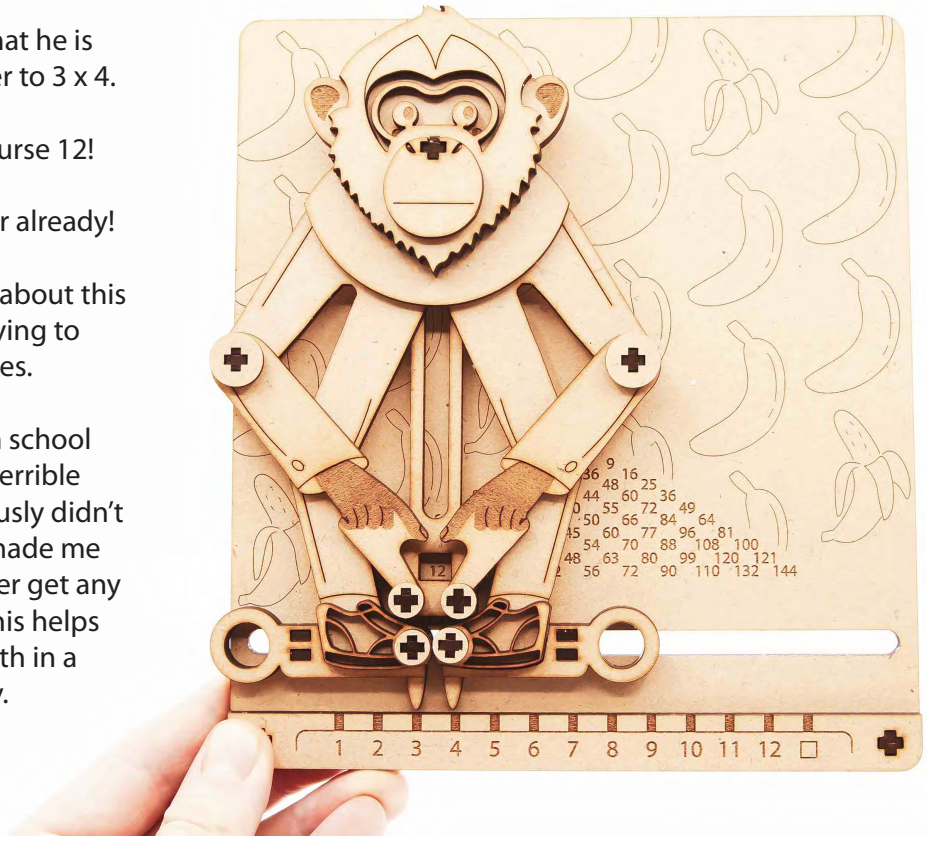
Now you can see that he is showing the answer to 3 x 4.

The answer is of course 12!

I am feeling smarter already!

I wish I had known about this when I was a kid trying to learn my times tables.

My math teacher in school was an absolutely terrible teacher who obviously didn't like his job which made me hate math and never get any good at it. I hope this helps someone enjoy math in a simple and fun way.





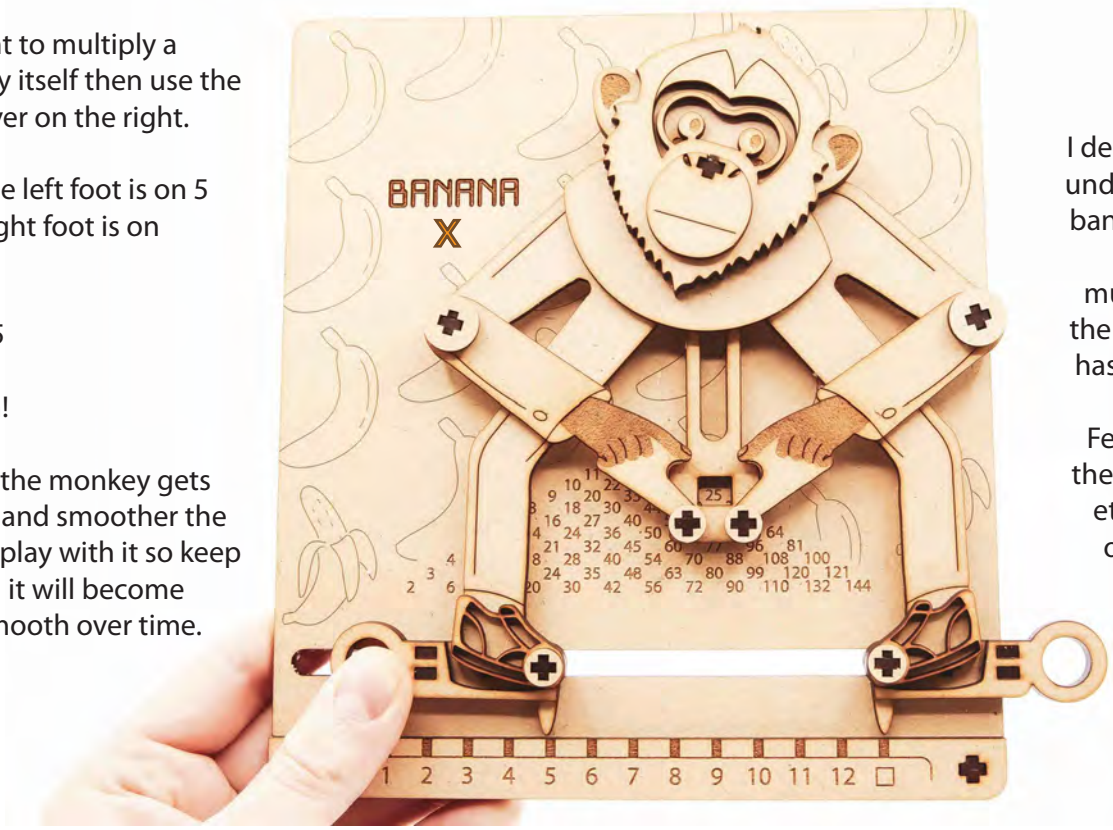
If you want to multiply a number by itself then use the 'square' over on the right.

So here the left foot is on 5 and the right foot is on squared...

So 5x5=25

Perfection!

I find that the monkey gets smoother and smoother the more you play with it so keep going and it will become buttery smooth over time.



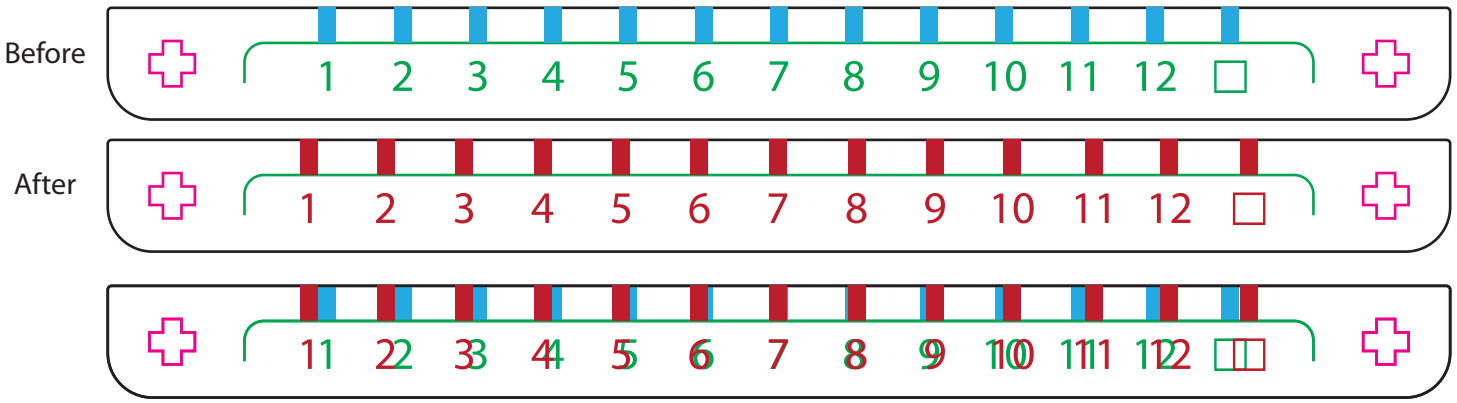
I decided to add an X underneath the word banana so you know this is for multiplication and the other back panel has a + for addition.

Feel free to delete the writing/bananas etc and add your own designs or decorations..

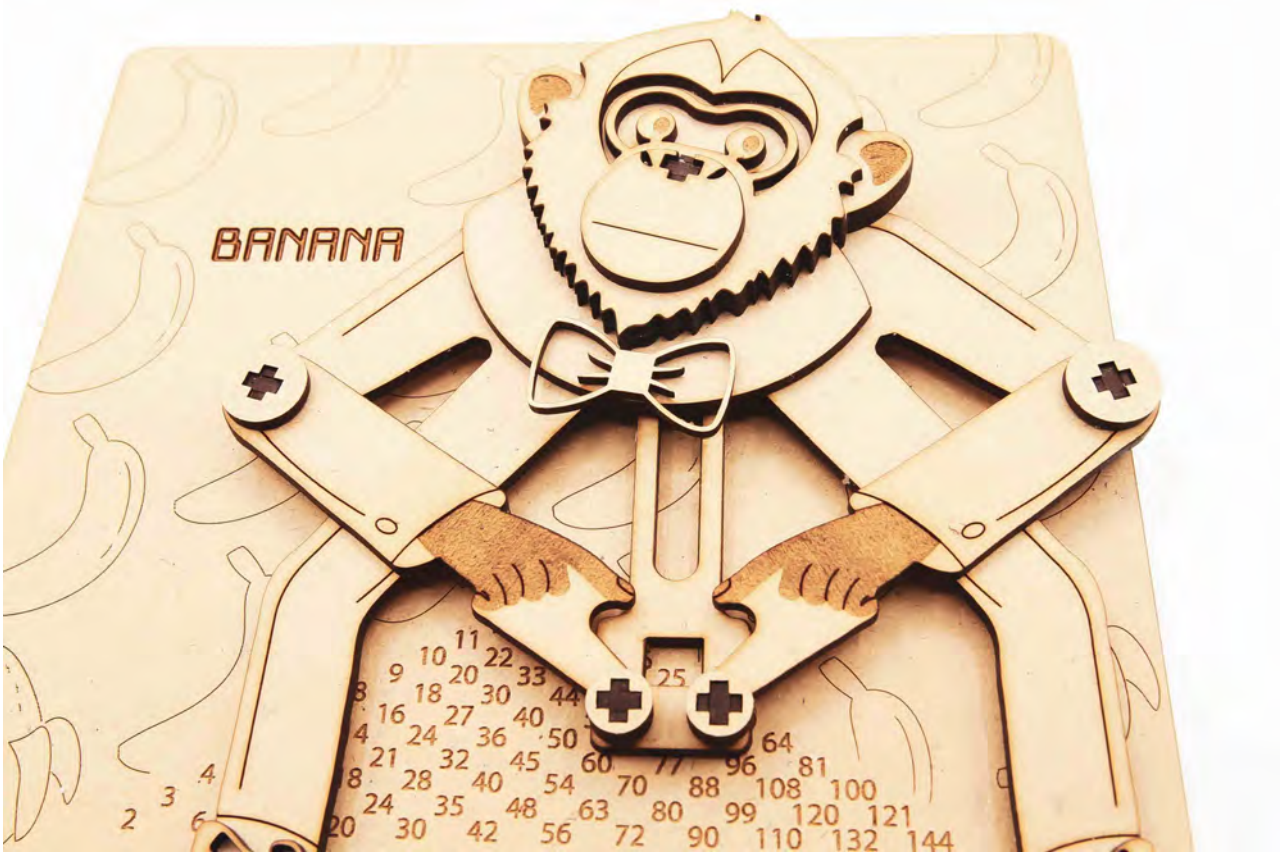
Get creative!

Troubleshooting

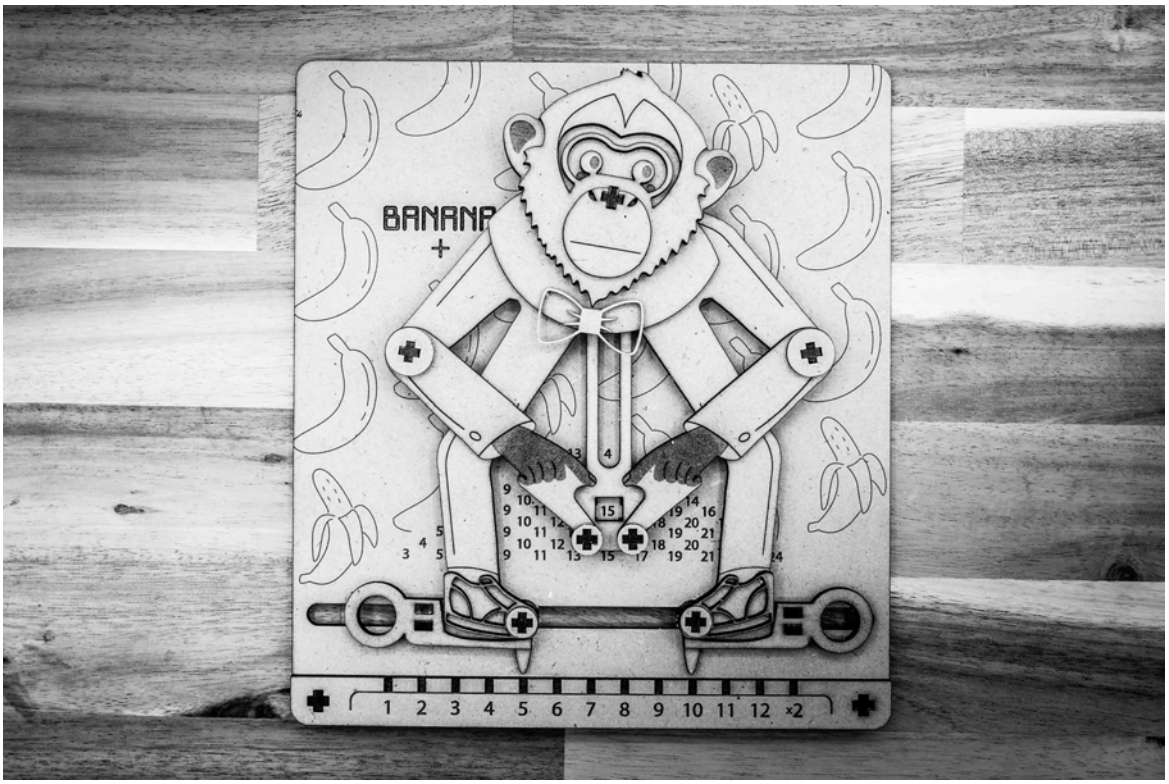
Tadaa! Originally the numbers were printed directly onto the backing board but I started thinking.... I wonder if there will be someone out there who the numbers just don't line up for... this could happen if there is too much play in your wood, probably if the pieces are too loose. The test cut file at the beginning should avoid this problem but just in case here is somethign you can do. Open up the files in some kind of editing software like Adobe Illustrator, Corel Draw etc and try and stretch out the numbers a little bit like this.



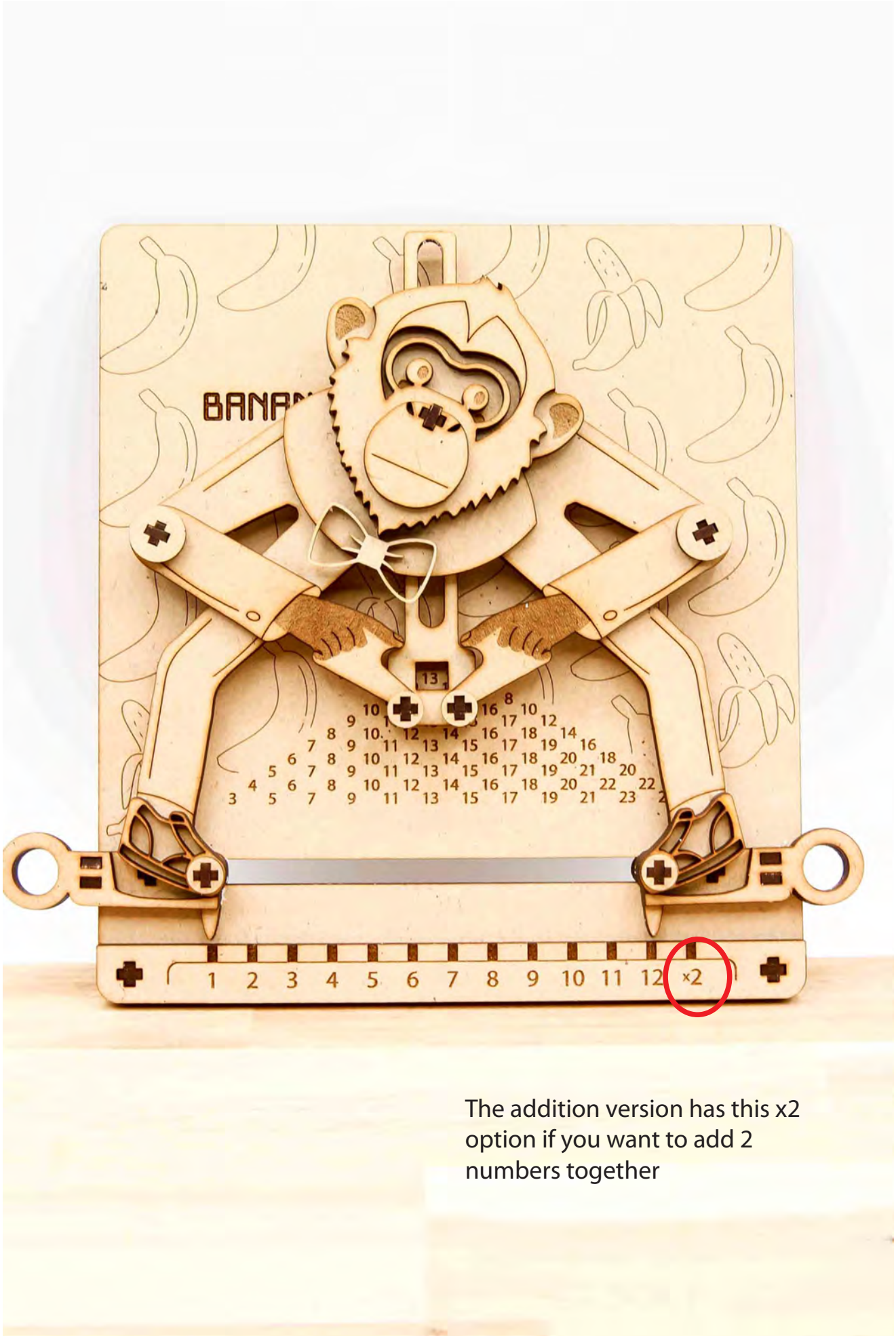
If you run into any problems please send me an email and I will try and troubleshoot it with you together!



I forgot the most important part! The bow-tie. If you are feeling dapper then just add a drop of glue in the middle and stick it on



And I think we are done! Here are some beauty shots. Feel free to paint and play around with it as much as you like! Just remember to please never share the digital files as I do this for fun. I don't want to stop making these kinds of files so let's all be nice people and look after one another.



The addition version has this x2 option if you want to add 2 numbers together



The multiplication version has this square (as in squaring a number) if you want to multiply 2 numbers together.